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APPLIED AI · MACHINE LEARNING · DEEP LEARNING ·
EXPLAINABLE AI · TIME-SERIES FORECASTING

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RESEARCH SIGNATURE

I design and ship AI systems that **predict, explain, and reveal** what happens inside learning models. My work spans INTERPRETABLE AND PHYSICS-INFORMED AI, TIME-SERIES FORECASTING, COMPUTER VISION, DIGITAL FORENSICS, AND LLMS, with **19 peer-reviewed publications** across CVPR, IEEE, ELSEVIER, AND SPRINGER. *One thread connects it all: making powerful models more understandable, measurable, and useful in the real world.*

RESEARCH EXPERTISE

ML and Deep Learning	LLMs, PyTorch, TensorFlow/Keras, scikit-learn, transformers, CNNs, RNNs/LSTMs, GNNs, KANs, hypernetworks, and diffusion models.
Time Series and Energy AI	Solar and load forecasting, building-energy modeling, physics-informed learning, temporal validation, and domain-constrained modeling.
Explainable and Human-Centered AI	Mechanistic interpretability, neuron-level analysis, non-interventional explainability, model auditing, trust, usability, and uncertainty communication.
Research Engineering	Data pipelines, benchmarking, ablation studies, statistical testing, failure analysis, APIs, testing, CI/CD, deployment, and reproducible workflows.

EDUCATION

Dec. 2025	PhD, Electrical and Computer Engineering Western University · London, Ontario, Canada Specialization: Applied Artificial Intelligence and Explainable AI Dissertation: <i>Neural Networks and Their Interpretability for Building Energy Modeling</i> . Supervisor: Dr. Katarina Grolinger .
Dec. 2019	MS, Software Engineering The University of Lahore · Lahore, Pakistan Thesis: <i>Image Denoising and Restoration Using Convolutional Neural Networks</i> Advisor: Dr. Muhammad Sultan Zia .
July 2018	BS, Software Engineering COMSATS University Islamabad · Pakistan Capstone: <i>A Comparative Analysis of Image Restoration Techniques</i> Advisor: Dr. Muzammil Behzad .

CURRENT APPOINTMENTS

Jan. 2026 – Present	Postdoctoral Fellow – Applied AI and Research Software Western University · London, Ontario, Canada Develop physics-informed AI, mechanistic interpretability, human-centered XAI, computer-vision, and digital-forensics systems. Designed and led Linear Lens , a non-interventional mechanistic method for explainable AI, verified across 45 datasets, 180 model analyses, 9,720 neuron records , and 36 recovery scenarios , and validated in a 400-participant human evaluation. Recipient of the Western Postdoctoral Fellowship , awarded competitively across the university. Supervisor: Dr. Katarina Grolinger
June 2023 – Present	Research Assistant – Music Data Science and Software Wilfrid Laurier University · Waterloo, Ontario, Canada

Co-develop **MidiPy**, open-source software for MIDI parsing, feature extraction, batch/segment analysis, visual analytics, PyPI distribution, and Streamlit deployment for health-science research.
Live application: midipy-online.streamlit.app

PEER-REVIEWED PUBLICATIONS

19 published or accepted. Titles, DOI labels, and record labels open the publication or venue page. Citation snapshot: Aug. 2026.

JOURNAL ARTICLES

- J1. **M. U. Danish** and K. Grolinger. “Leveraging Hypernetworks and Learnable Kernels for Consumer Energy Forecasting Across Diverse Consumer Types.” **IEEE Transactions on Power Delivery**, 40(1), 75–87, 2024. DOI: [10.1109/TPWRD.2024.3486010](https://doi.org/10.1109/TPWRD.2024.3486010).
IF 4.7 · CITESCORE 11.2 · CITATIONS 19
- J2. **M. U. Danish**, K. Ali, K. Siddiqui, and K. Grolinger. “Physics-Guided Memory Network for Building Energy Modeling.” **Energy and AI**, 21, 100538, 2025. DOI: [10.1016/j.egyai.2025.100538](https://doi.org/10.1016/j.egyai.2025.100538).
IF 9.7 · CITESCORE 16.3 · CITATIONS 5
- J3. M. Aziz, U. Rehman, **M. U. Danish**, and K. Grolinger. “Global-Local Image Perceptual Score (GLIPS): Evaluating Photorealistic Quality of AI-Generated Images.” **IEEE Transactions on Human-Machine Systems**, 55(2), 223–233, 2025. DOI: [10.1109/THMS.2025.3527397](https://doi.org/10.1109/THMS.2025.3527397).
IF 4.4 · CITESCORE 10.6 · CITATIONS 35
- J4. **M. U. Danish**, M. Aziz, U. Rehman, and K. Grolinger. “Linear Lens: A Human-Centered, Non-Interventional Mechanistic Approach to Explainable AI.” **Machine Learning with Applications**, 24, 100897, 2026. DOI: [10.1016/j.mlwa.2026.100897](https://doi.org/10.1016/j.mlwa.2026.100897).
IF 6.1 · CITESCORE 12.6 · CITATIONS 0
- J5. **M. U. Danish** and K. Grolinger. “Kolmogorov–Arnold Recurrent Network for Short-Term Load Forecasting Across Diverse Consumers.” **Energy Reports**, 13, 713–727, 2025. DOI: [10.1016/j.egyr.2024.12.038](https://doi.org/10.1016/j.egyr.2024.12.038).
IF 6.6 · CITESCORE 15.3 · CITATIONS 56
- J6. M. Buwaneswaran, **M. U. Danish**, and K. Grolinger. “Conditional Denoising Diffusion Model for Missing Value Imputation in Smart Meter Data.” **Engineering Applications of Artificial Intelligence**, 181, 115403, 2026. **In press**.
IF 9.0 · CITESCORE 11.7 · CITATIONS 0
- J7. M. Aziz, U. Rehman, **M. U. Danish**, S. Ali, and A. Z. Abbasi. “Towards a Unified Evaluation Framework: Integrating Human Perception and Metrics for AI-Generated Images.” **Multimedia Systems**, 31(3), 180, 2025. DOI: [10.1007/s00530-025-01769-7](https://doi.org/10.1007/s00530-025-01769-7).
IF 2.8 · CITESCORE 6.7 · CITATIONS 21
- J8. **M. U. Danish**, U. Rehman, and K. Grolinger. “Monotone Delta: An Order-Theoretic Tournament Graph Approach for Internal Consistency Assessment.” **Frontiers in Applied Mathematics and Statistics**, 12, 1687194, 2026. DOI: [10.3389/fams.2026.1687194](https://doi.org/10.3389/fams.2026.1687194).
IF 2.0 · CITESCORE 3.6 · CITATIONS 0
- J9. D. Kogutec and **M. U. Danish**. “MidiPy: A Python Application for Processing MIDI Data in Improvised Active Music Therapy.” **Journal of Creative Music Systems**, 2026. **Accepted**. Live software.
- J10. D. Kogutec, L. Carroll, **M. U. Danish**, and C. Lu. “Dispersy Measures Within Sessions During Improvised Active Music Therapy with Clients with Parkinson's Disease.” **Approaches: An Interdisciplinary Journal of Music Therapy**, 2026. **Accepted for publication**.

CONFERENCE PROCEEDINGS

- C1. D. Tissera, O. Awadallah, **M. U. Danish**, A. Sadhu, and K. Grolinger. “Any-Class Presence Likelihood for Robust Multi-Label Classification with Abundant Negative Data.” **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**, 2026. CVPR record.
- C2. **M. U. Danish**, M. Aziz, U. Rehman, and K. Grolinger. “Cognitive Effects of Textual, Graphical, and Interactive Explanations in Explainable Artificial Intelligence.” **52nd Annual Conference of the IEEE Industrial Electronics Society (IECON 2026)**, 2026. **Accepted**.
- C3. **M. U. Danish**, M. Aziz, U. Rehman, A. Zecevic, and K. Grolinger. “Human-Centered Explainable Artificial Intelligence and the Effects of Explanation Modalities on Trust, Understanding, and Decision Making.” **IECON 2026**, 2026. **Accepted**.

- C4. **M. U. Danish**, M. Buwaneswaran, T. Fonseka, and K. Grolinger. “Graph Attention Convolutional U-Net: A Semantic Segmentation Model for Identifying Flooded Areas.” **IECON**, 2024. DOI: 10.1109/IECON55916.2024.10905482.
- C5. **M. U. Danish**, M. Noorchenarboo, A. Narayan, and K. Grolinger. “ChebyRegNet: An Unsupervised Deep Learning Technique for Deformable Medical Image Registration.” **IECON**, 2024. DOI: 10.1109/IECON55916.2024.10905705.
- C6. M. Aziz, **M. U. Danish**, and U. Rehman. “Distributional Feature Separability for Explainability of Neural Networks for Financial Credit Evaluation.” **IEEE AIXSET**, 2025. **Best Paper Award**. DOI: 10.1109/AIXSET65682.2025.00024.
- C7. **M. U. Danish**, A. A. Yousaf, M. Aziz, K. Grolinger, and U. Rehman. “RootSolver: A Cognitive-Inspired Approach to Polynomial Root-Finding with Multiplicity Detection.” **17th International Conference on Advanced Computational Intelligence**, 2025. DOI: 10.1109/ICACI65340.2025.11096224.
- C8. A. A. Yousaf, M. Aziz, **M. U. Danish**, S. A. Safi, and U. Rehman. “Statistically-Guided Kernel Regression: Embedding Feature Relevance into Kernel Learning.” **2nd International Conference on Advanced Machine Learning and Data Science**, 2026. **Accepted**.
- C9. M. Aziz, **M. U. Danish**, F. Iqbal, and U. Rehman. “Finite-Difference Approximation for Explaining Neural Network Predictions in Financial Credit Evaluation.” **International Conference on Advanced Machine Learning and Data Science**, 2025. DOI: 10.1109/AMLDS63918.2025.11159472.

PREVIOUS RESEARCH, TEACHING, AND PROFESSIONAL EXPERIENCE

- Jan. 2026 –
August. 2026 **Postgraduate Researcher – Human-Centered AI and HCI**
Western University · London, Ontario, Canada
Design human-AI studies evaluating explanation modality, trust, understanding, cognitive load, decision quality, appropriate reliance, usability, and uncertainty communication.
Focus: study design, ethics, validated measures, statistical analysis, and reproducible reporting.
- Jan. 2026 –
Oct. 2026 **Research Engineer – AI and Digital Forensics**
Zayed University · Remote / United Arab Emirates
Develop reproducible Python workflows for digital-forensics research, including data processing, benchmarking, robustness, attribution, explainability, and failure analysis across Windows, Linux, and memory artifacts.
Scale: approximately **2.6 million malware records**.
- Jan. 2023 –
Dec. 2025 **Graduate Research Assistant – Machine Learning**
Western University · London, Ontario, Canada
Built end-to-end ML systems for energy forecasting, building-energy modeling, missing-value imputation, computer vision, and XAI; developed first-author methods using hypernetworks, KAN-based recurrent models, and physics-guided memory networks.
- June 2022 –
Dec. 2022 **Research Assistant – Computer Vision**
Gwangju Institute of Science and Technology · Gwangju, South Korea
Implemented and benchmarked deep-learning methods for image restoration, deformable medical-image registration, and semantic segmentation with controlled evaluation.
- Jan. 2023 –
Dec. 2025 **Graduate / Lead Teaching Assistant**
Western University · London, Ontario, Canada
Completed **713 contracted hours** across Programming Fundamentals for Engineers, Database Management Systems, Software Design, and Software Engineering Design I; served as Lead TA for SE 3309 in Fall 2024.
- Jan. 2019 –
Dec. 2021 **Lecturer**
The University of Lahore · Lahore, Pakistan
Taught machine learning and statistics, designed assessments, and supervised applied student projects.

SELECTED PROJECTS AND RESEARCH SOFTWARE

Core-Norm	Robust, bounded, invertible preprocessing for numeric ML; 12,150 evaluations , 25 verification datasets , scikit-learn integration, automated tests, and CI.
Linear Lens	Peer-reviewed, non-interventional mechanistic interpretability; 45 datasets , 180 trained-model analyses , neuron-level diagnostics, and human-centered evaluation.
OpenMetricLab	Local-first evaluation for regression, classification, segmentation, and image registration with reproducible metrics and model comparison.
MidiPy	MIDI parsing, feature extraction, batch/segment analysis, visual analytics, PyPI packaging, and Streamlit deployment.
Additional public systems	AlgoLens , StructureDoctor , EdgeCaseForge , AlgoTwin , RefactorDS , MatMulX , MatrixMorph , SparseScope , and MatrixTrace . 13 public systems total.

AWARDS AND FELLOWSHIPS

2026	Western Postdoctoral Fellowship , Western University
2025	Best Paper Award , IEEE AIXSET
2024	Western Graduate Fellowship , Western University
2023	Western Graduate Research Scholarship , Western University
2022	Korean Government Scholarship

PROFESSIONAL SERVICE

Journal reviewer	Engineering Applications of Artificial Intelligence ; IEEE Transactions on Industrial Informatics ; IEEE Transactions on Artificial Intelligence ; Energy and AI ; Applied Energy .
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TECHNICAL SKILLS

Programming and Data	Python, SQL/MySQL, data structures and algorithms, MATLAB, Bash, NumPy, SciPy, pandas, Dask.
ML and Deep Learning	PyTorch, TensorFlow/Keras, scikit-learn, transformers, CNNs, RNNs/LSTMs, GNNs, KANs, hypernetworks, diffusion models.
NLP, IR, and GenAI	spaCy, NLTK, RAG architectures, LangChain, LangGraph, agentic frameworks, tool-using systems, and orchestration patterns.
Time Series and Energy	Solar forecasting, load forecasting, building-energy modeling, and physics-informed AI.
Cloud and Engineering	AWS, Azure, Git/GitHub, Linux, Jupyter, Conda, REST APIs, CI/CD, pytest, PyPI, Streamlit, and Power BI.
Languages	English (professional fluency) · Urdu (native).